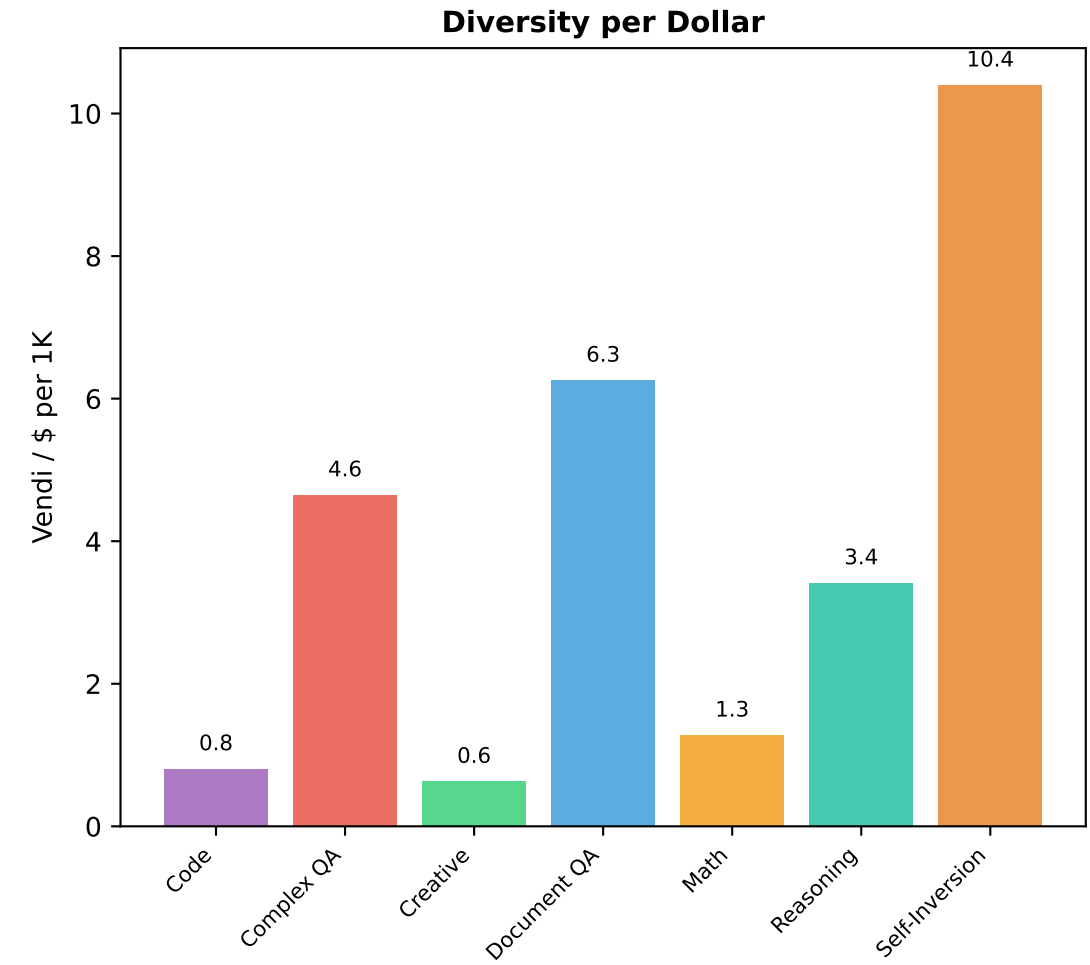
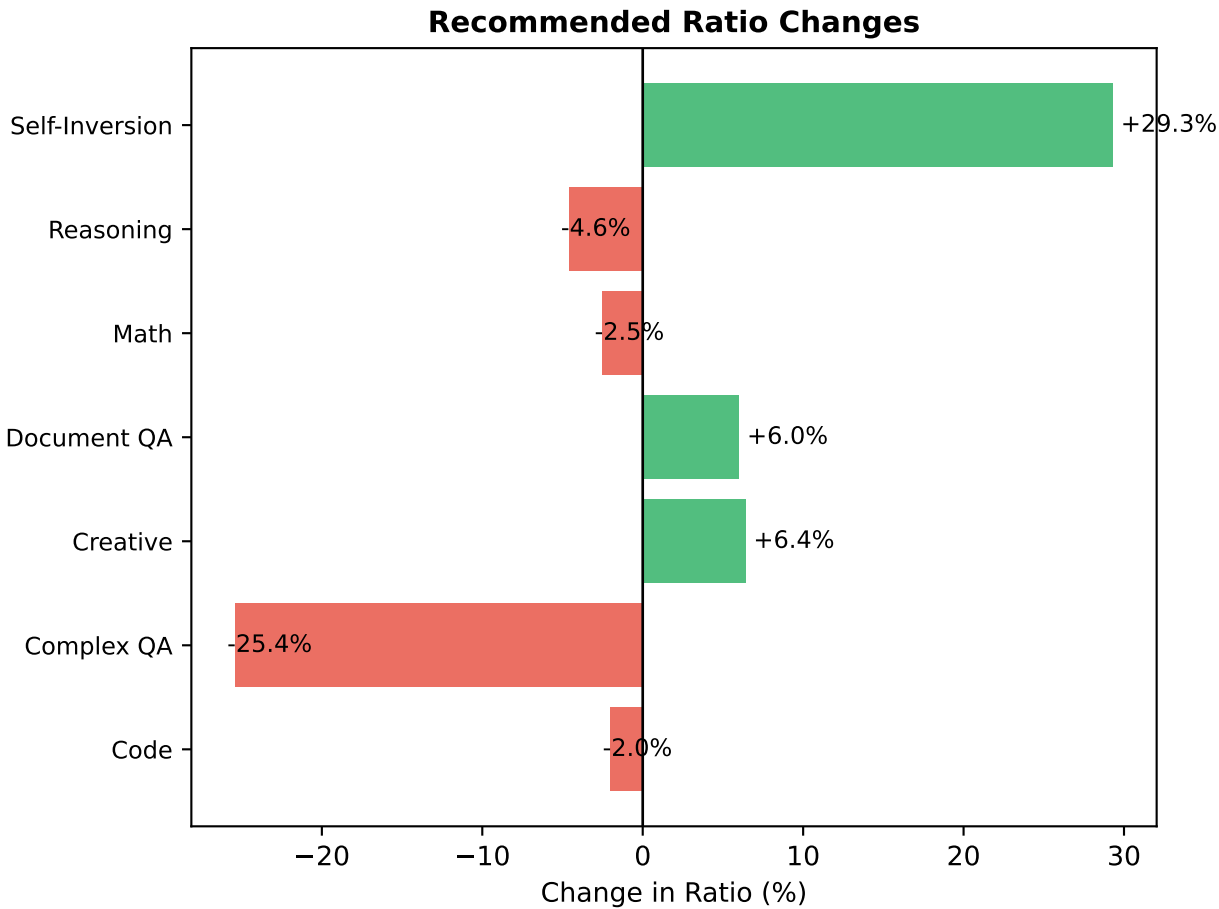
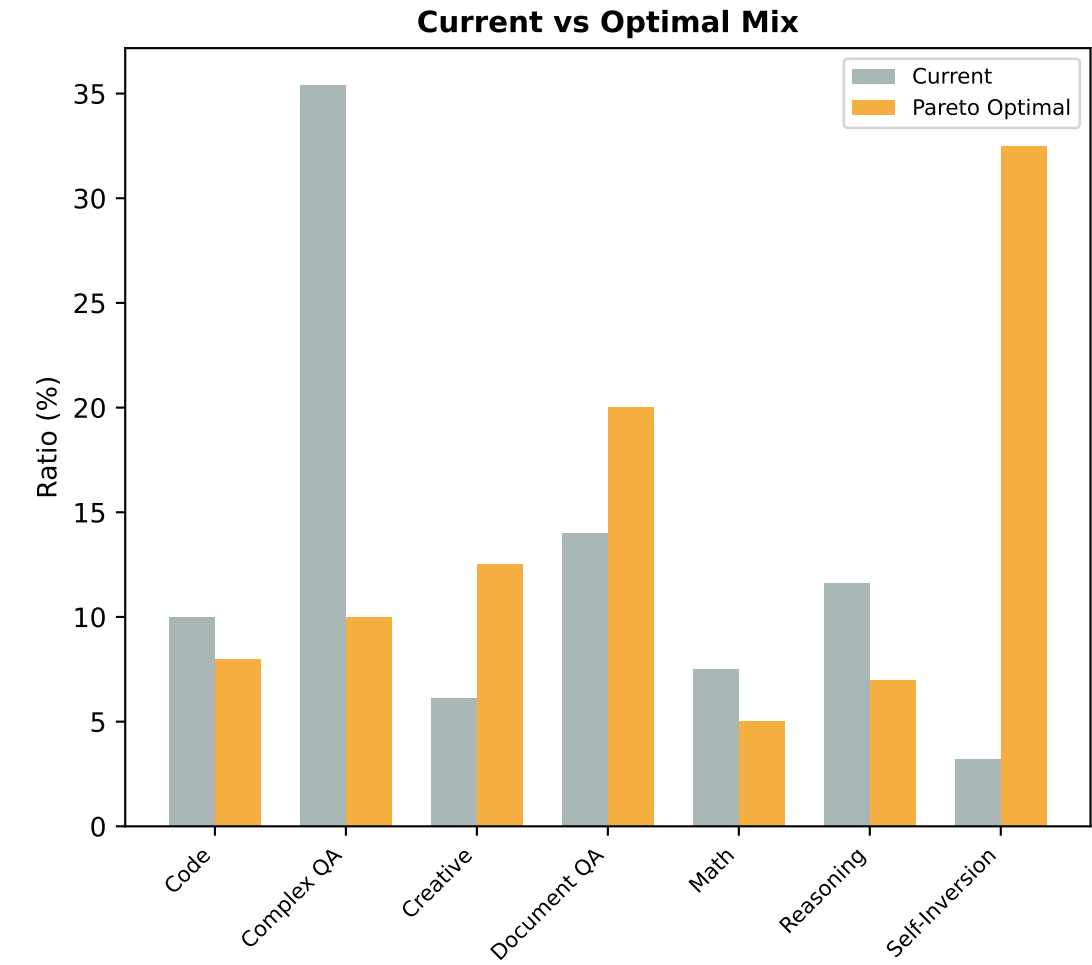
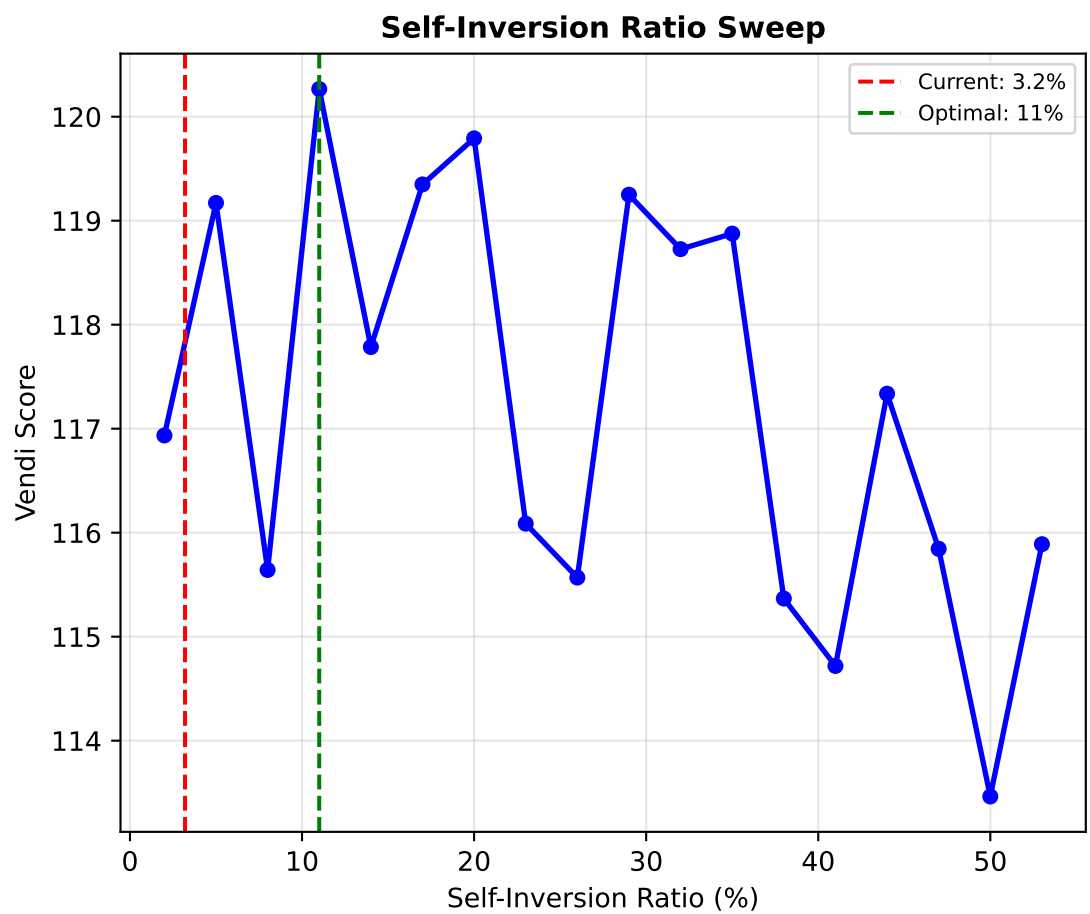
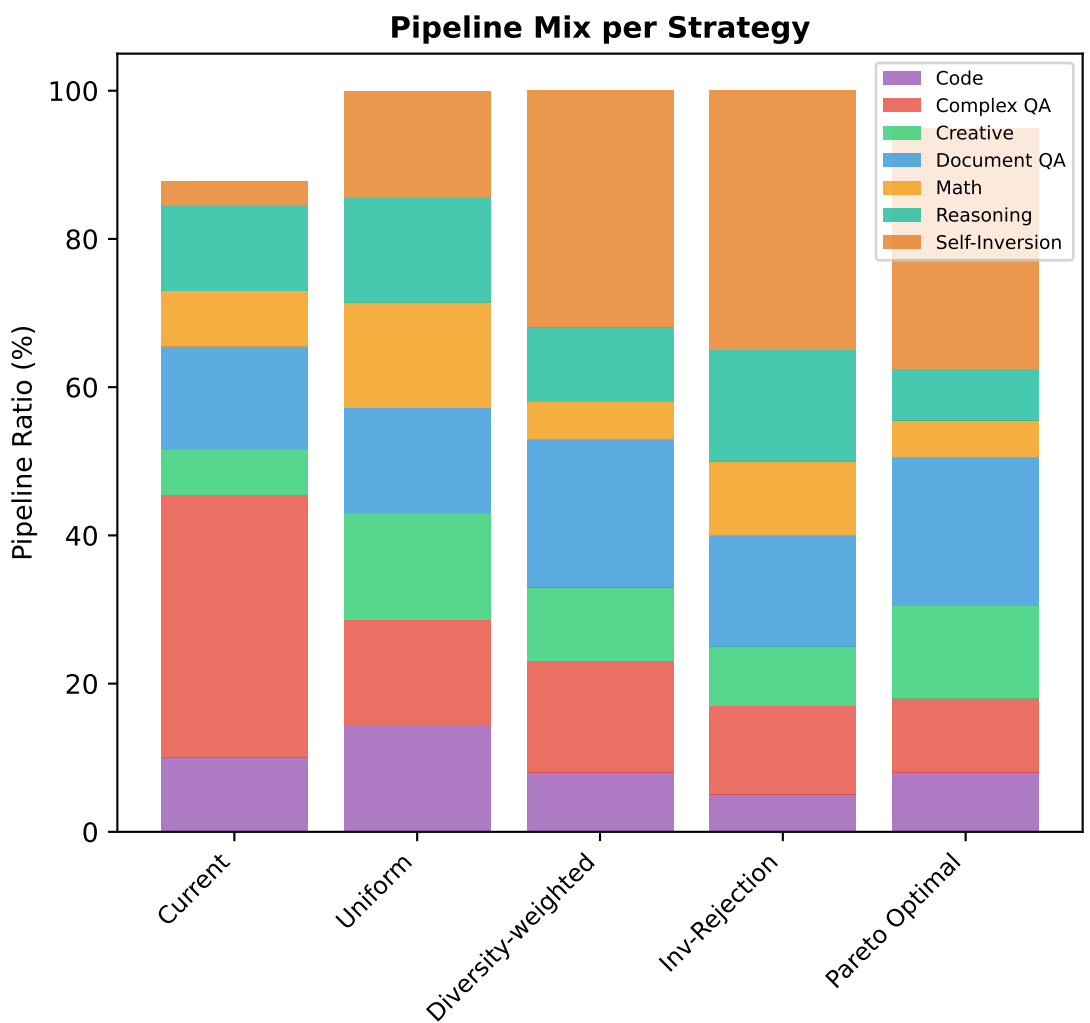
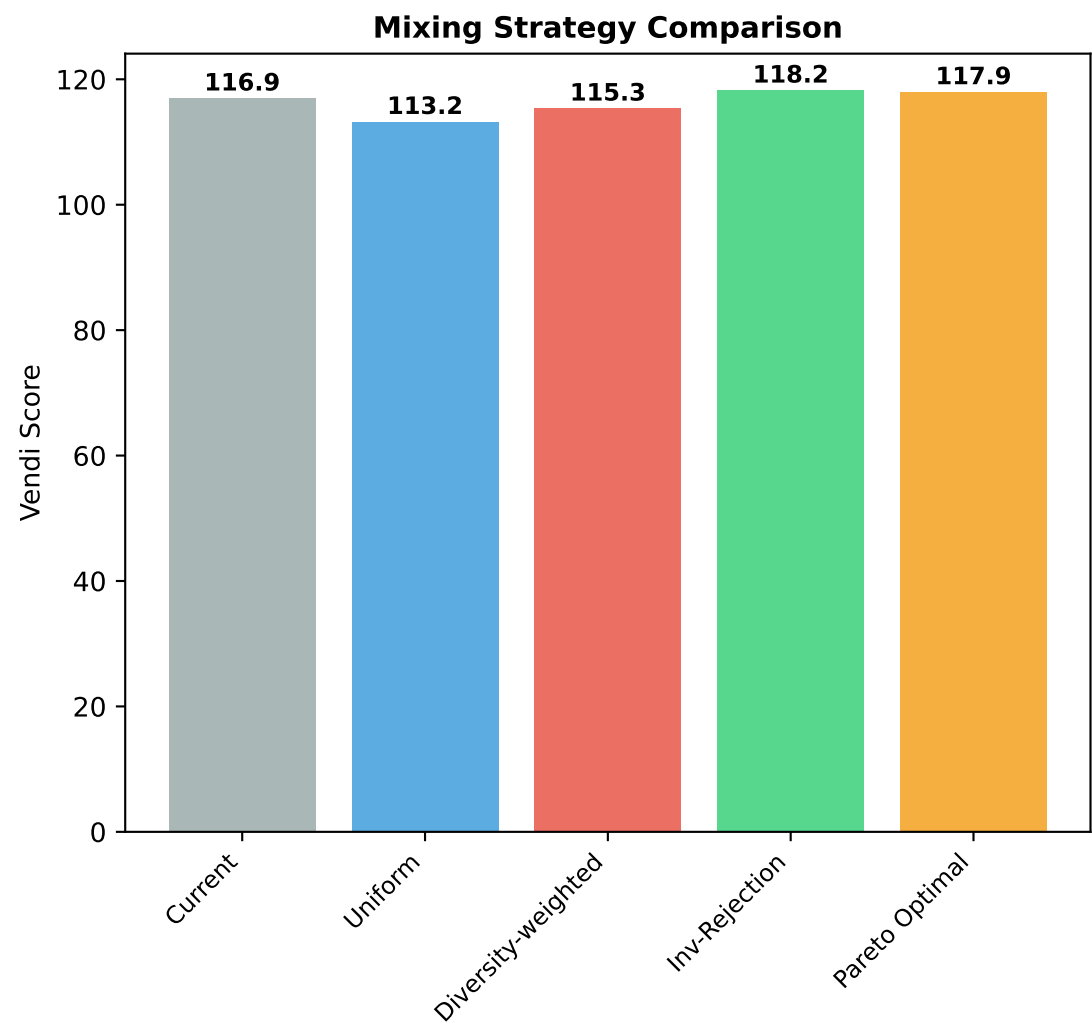


Fig 182: Optimal Pipeline Mixing Simulation



Recommended Sample Allocation by Budget

Budget	Code	Complex	Creative	Document	Math	Reasonin	Self-Inv
500	40	50	62	100	25	35	162
1,000	80	100	125	200	50	70	325
5,000	400	500	625	1,000	250	350	1,625
10,000	800	1,000	1,250	2,000	500	700	3,250
50,000	4,000	5,000	6,250	10,000	2,500	3,500	16,250
100,000	8,000	10,000	12,500	20,000	5,000	7,000	32,500

Mixing Optimization Summary

Best strategy: Inv-Rejection
Vendi = 118.2

Current Vendi: 116.9
Best achievable: 120.3
Improvement: +2.9%

Key changes recommended:
Self-Inversion: 3.2% -> 11%
Creative: 6.1% -> 12.5%
Complex QA: 35.4% -> 10%
Document QA: 14% -> 20%

Most cost-efficient: Self-Inversion
(10.4 Vendi/\$)
Least cost-efficient: Code
(0.8 Vendi/\$)